



13

More About Solid Shapes

Tune In



Five children dream of becoming solid shape detectives. To do so, they must solve the mystery correctly. They would appreciate it if you could match the mysteries to their respective solid shape so that they could fulfil their dreams.



There is only one curved face and no vertices of the shape I am looking for, but it has the same edges and flat faces.



I am looking for a shape that has twelve edges, six faces, and eight vertices, but all the faces look the same.



In shape I am looking for, there are twelve edges, six faces, and eight vertices with the opposite faces equal in appearance.

Cube

Cylinder

Cone

Cuboid

Sphere

There should be no edges or flat surfaces in the shape I am looking for. It has only one curved face.



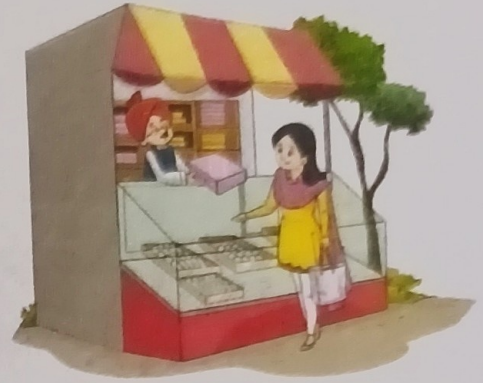
The shape I am looking for must have a similar number of edges, flat faces, curved surfaces, and vertices. People wear objects made from this shape at parties.



NETS OF SOLIDS

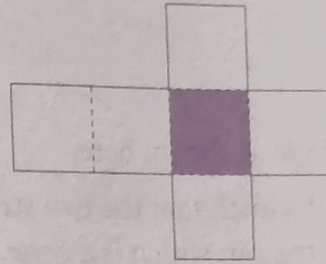
Kesha and her mother went to Shudh Sweets Shop to buy sweets for Diwali. Kesha noticed that the cardboard boxes used for packing the sweets were made from specially cut-out sheets. By folding the sheets along certain lines, the packer was able to make the boxes easily.

A **net** is a pattern that you can cut and fold to make a model of a solid shape.

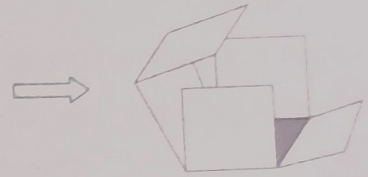


Net of a Cube

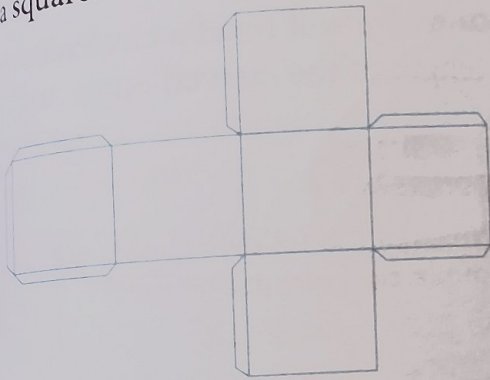
If we copy and cut the first picture along the outside edges and then fold along the dotted lines we will get a cube. We can see that a cube has 6 faces, each of which is a square.



net of a cube



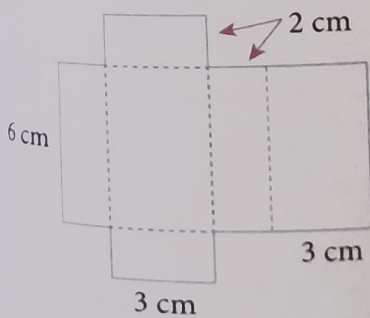
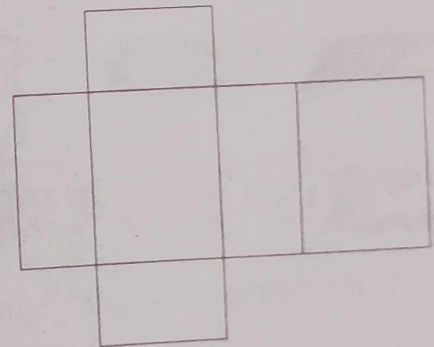
cube



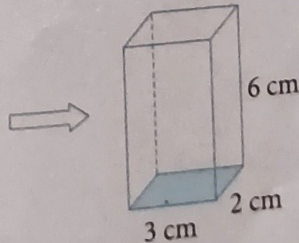
The picture that we see here is also the net of a cube. We can see small flaps along some edges. When we are making the cube, we can apply glue on these flaps and stick them on the overlapping face so that the cube will retain the shape and will be a closed solid.

Net of a Cuboid

Have you opened out the box in which your birthday cake was packed and brought home? It probably looked like the net given here. This net, when folded, turns into a cuboid.



net of a cuboid

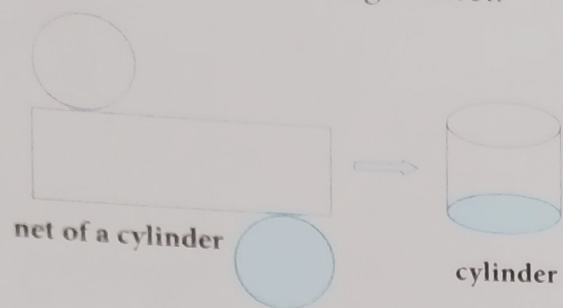


cuboid

The net of a cuboid has six rectangles. As the opposite faces of a cuboid are of the same size, the net has three pairs of equal rectangles.

Net of a Cylinder

Take a rectangular sheet of paper and roll it such that the two edges meet.

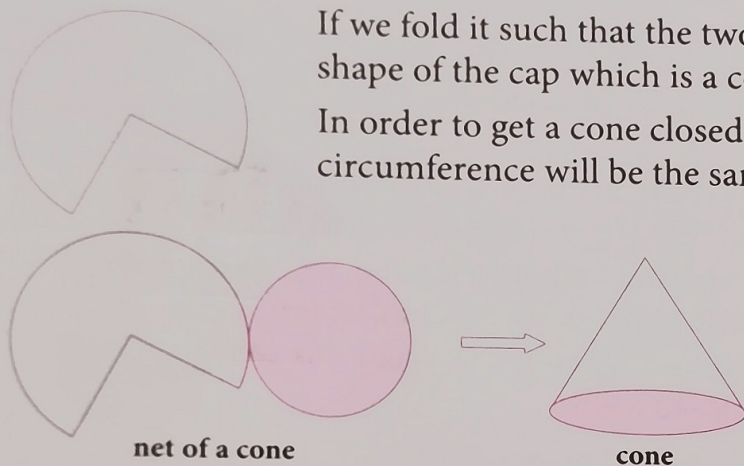


We get a cylinder with both the ends open. We need two circular sheets of paper to close the cylinder at both ends. The net of a closed cylinder will look like the one given on the left. The circles can be placed at any point on the sides. The circumference of the circles should be equal to the length of the sides to which they are attached.

Net of a Cone

Open out a birthday cap.

We will get a portion of a circle as shown here.

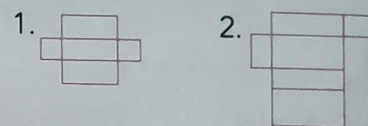


If we fold it such that the two straight edges meet, we will get the shape of the cap which is a cone.

In order to get a cone closed at the bottom, we will need a circle whose circumference will be the same as the length of the curved edge.

Let's Try! 1

Which of the following nets will form a closed cuboid?



Exercise 13A

Go to MyInspire and attempt questions 1 and 2.

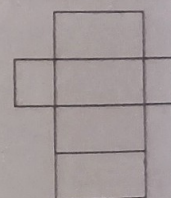
3. Circle the name of the solid figure the given net makes.

Cuboid

Cylinder

Cone

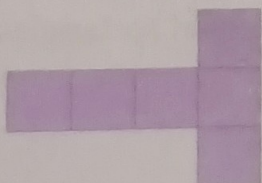
Cube



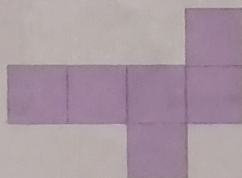
Application Based Questions

4. Which of the following nets will fold into a cube?

a.

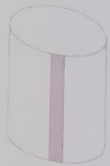


b.



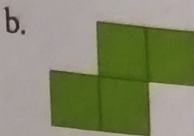
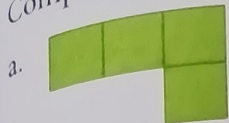
c.





a. We need
cylinder at
all look like
e placed
ce of the
e sides to

Complete the following nets by adding two squares.



DRAWING SOLID SHAPES

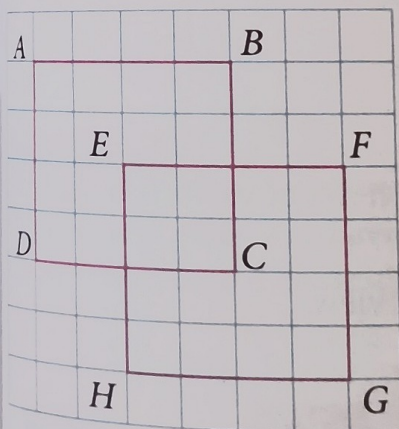
Let us learn to draw some solid shapes.

Cube

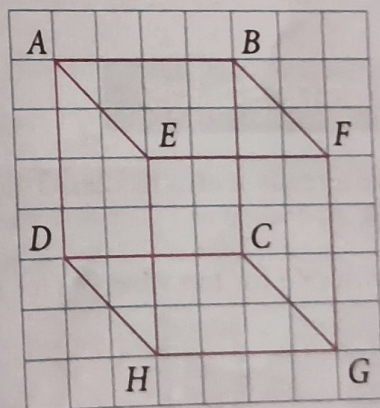
A cube has 6 faces all of which are squares. Since we are drawing the picture of a solid object on a flat sheet of paper, we will not be able to show all the faces as squares. But if we look at our drawing we will be able to recognise that we have drawn a cube. Now let us draw a cube.

Step 1: Take a sheet of paper with square grid.

Step 2: Draw two squares – the second one slightly below and to the right of the first square. Name the corners $ABCD$ and $EFGH$.



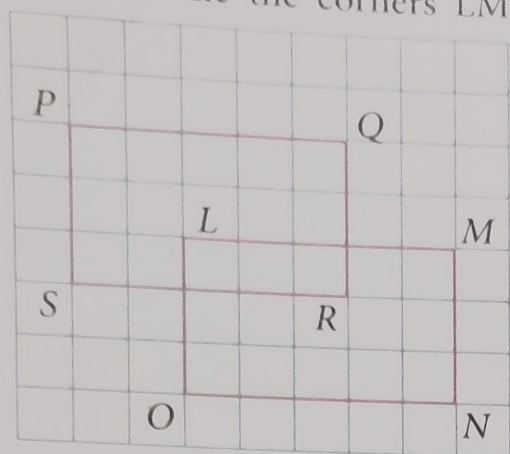
Step 3: Join the corners AE , BF , CG , and DH .



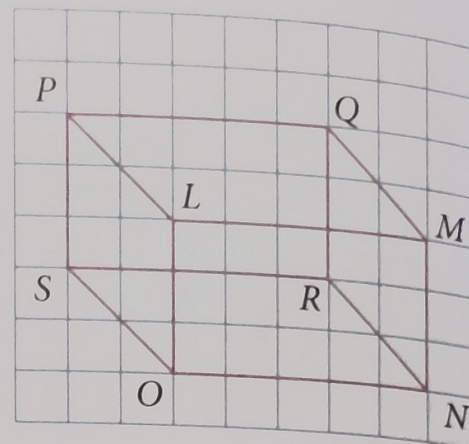
Cuboid

A cuboid has 6 faces and the opposite faces are identical. We can draw a cuboid in the same way as we draw a cube.

Step 1: Draw two identical rectangles as shown below. Name the corners LMNO and PQRS.



Step 2: Join the corners PL, SO, QM, and RN.



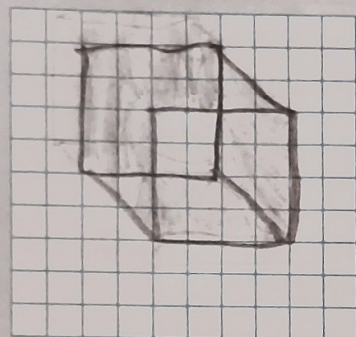
Go to MyInspire for



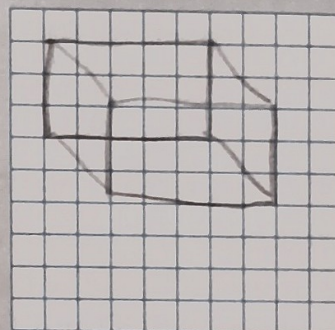
Exercise 13B

Draw the following solid shapes.

1. Cube



2. Cuboid



VIEWS OF SOLIDS

We have already learnt in Class 3 about the top, front, and side views of various objects around us.

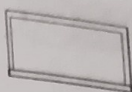
Mark with (✓) for top view, (✕) for side view, and (☆) for front view.



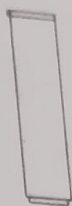
Now let us draw the different views of the given object.



Front



Top



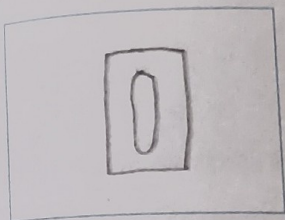
Side

Go to MyInspire for

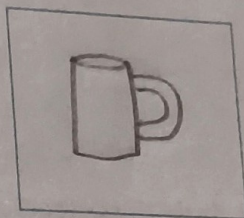
Exercise 13C

Go to MyInspire and attempt question 1.

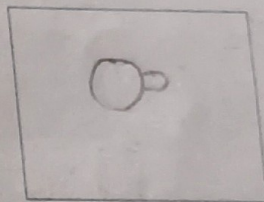
2. Draw the different views of a teacup with a handle.



Front (handle facing you)



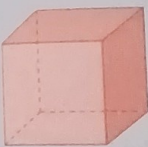
Side



Top

3. Draw the top, front, and side views of the solid shapes given below.

a.



b.



c.



d.



Let's Review

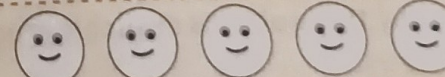


Come here after you have completed the chapter. Colour 1 to 5 😊 to show how well you understand a topic. (5 😊 = Very well; 1 😊 = Not at all)

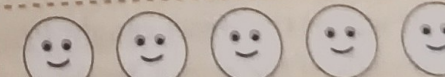
Nets of Solids



Drawing Solid Shapes



Views of Solids



Revision Exercises



Go to MyInspire for

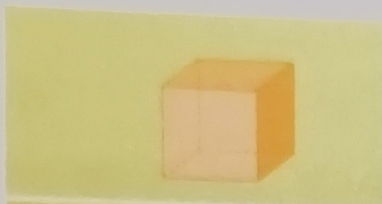


Start Up

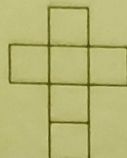


1. Circle the correct net for each solid.

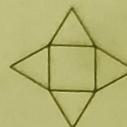
a.



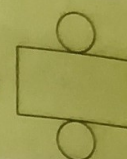
(i)



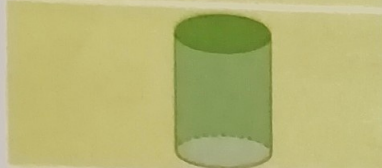
(ii)



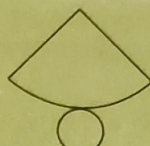
(iii)



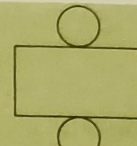
b.



(i)



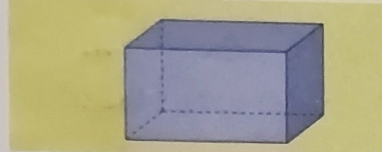
(ii)



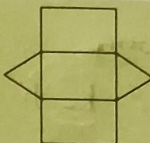
(iii)



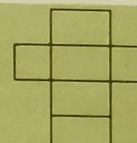
c.



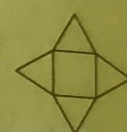
(i)



(ii)

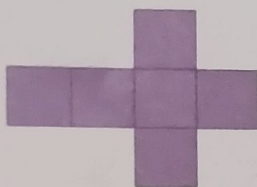


(iii)



2. Identify the following nets and circle the correct answer.

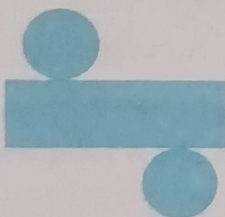
a. cube/cuboid/cone/cylinder



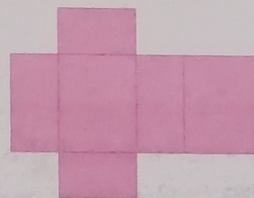
b. cube/cuboid/cone/cylinder



c. cube/cuboid/cone/cylinder



d. cube/cuboid/cone/cylinder



Level Up



1. Take any three objects available around you and draw their top, front, and side views.

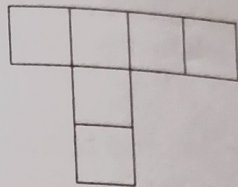
2. A house across the street caught Priyal and Rubina's attention. While Rubina could see two walls of the house, Priyal could only see one. Why is this so? Explain your answer.

HOTS Questions

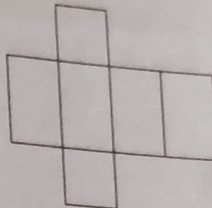


Which of the following nets will not make a cube?

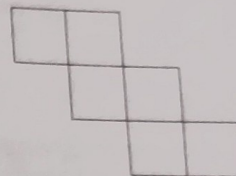
2.



3.



4.



Everyday Maths



Nets are used in many real-life situations. Sometimes birthday caps are sold in the form of nets. You glue the ends together to make the caps. Bakeries have nets of cuboids in various sizes for packaging their food items.

List objects you come across in your daily life that are in the shape of cubes or cuboids. Draw the shape of the object and their nets in your notebooks.

Cross-curricular Connect



(EVS) What do you do with waste in your home? Do you reuse any waste materials?

Discuss with your friends the importance of reducing garbage.

Use old newspapers to make paper bags. Follow the idea of nets you have learnt in this lesson to make the bags.

This will help you reduce wastage of materials and reuse things creatively.